Think about the situation 2 (Multiplication of a fraction using the operator 'of')

Kannan has 30 beads and Kanmani has one sixth **of** it. How many beads does Kanmani have?

Solution The number of beads that Kanmani has = $\frac{1}{6}$ **of** 30 beads

$= \frac{1}{6} \times 30 = \frac{30}{6}$

$= 5$ beads

Think about the situation 3 (Multiplication of a fraction by another fraction)

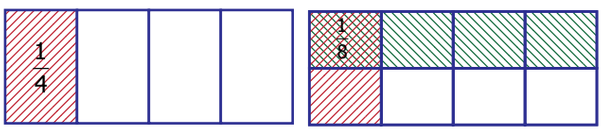
Sunitha bought three $\frac{1}{4}\ kg$ sweet packets for her three friends from a sweet stall. But 6 of her friends had come to her home. So she decided to divide each $\frac{1}{4} kg$ sweet packets into halves. If she has done in that way, what would be the weight of the sweet packet that each one of her friend will receive?

Solution The weight of the sweet packets that each one of her friends will receive } = half **of** $\frac{1}{4} kg$

$= \frac{1}{2}$ **of** $\frac{1}{4}\ kg$

$= \frac{1}{2} \times \frac{1}{4}$

The product $\frac{1}{2}$ and $\frac{1}{4}$ means $\frac{1}{2}$ **of** $\frac{1}{4}$. This can be illustrated as follows. We shade 1 part out of 4 parts which represents $\frac{1}{4}$. Now divide this horizontally into 2 equal parts and shade one part of it.



Here, the double shaded part represents the product $\frac{1}{8}$ and it is got by finding the product of the numerators and the product of denominators as follows:

$$\frac{1}{2} \times \frac{1}{4} = \frac{1 \times 1}{2 \times 4} = \frac{1}{8}$$

Example 16 Maruthu, a milk man has 4 bottles of milk each containing $1\frac{1}{2}$ litres. How much milk does he have in all?

Solution Since Maruthu has 4 bottles of milk and each containing $1\frac{1}{2}$ litres, he has 4 times of $1\frac{1}{2}$ litres of milk.

$$1\frac{1}{2} \times 4 = \left(1 + \frac{1}{2}\right) \times 4 = 4 + \frac{4}{2}$$

$= 4 + 2 = 6$ litres.

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